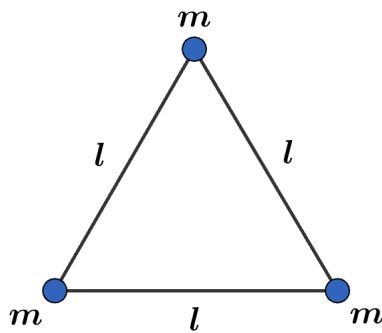
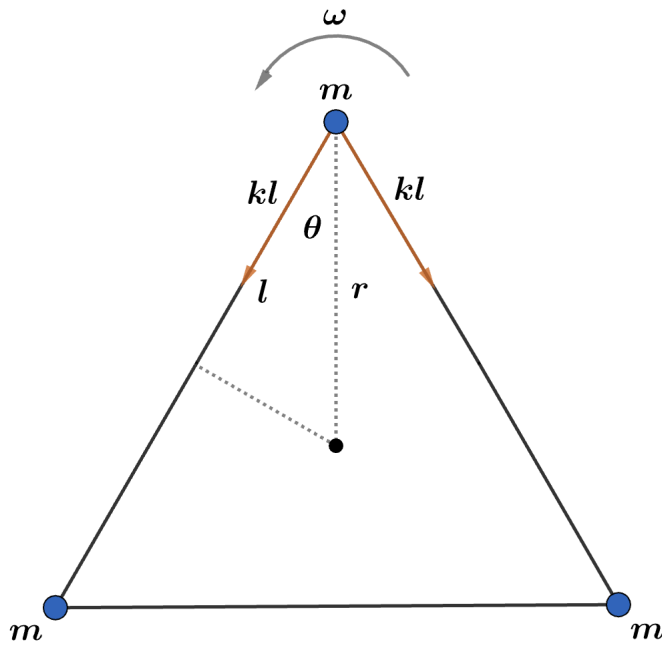


2012 F=ma Exam: Problem 24

Kevin S. Huang



When the assembly is rotated, the spring forces on the mass provide the centripetal acceleration. Since the springs are stretched by distance l , the spring forces have magnitude kl .



Applying Newton's 2nd law,

$$2kl \cos \theta = m\omega^2 r$$

From the figure, we have $r \cos \theta = l$ so

$$2kl \cos \theta = m\omega^2 \left(\frac{l}{\cos \theta} \right)$$

$$k = \frac{m\omega^2}{2 \cos^2 \theta}$$

Since we have an equilateral triangle, $\theta = \pi/6$ and $\cos \theta = \sqrt{3}/2$. Finally,

$$k = \frac{2m\omega^2}{3}$$

so the answer is $\boxed{\text{C}}$.